- https://medium.com/@hunkcool1991/uart-protocol-a-detailed-guide-7926d426581c
  - serial communication protocol
  - No shared clock line, means it is very strict in timing. What that mean tho? It means everything depends on the **baud rate**
  - Because there is no shared clock, each component works independent of the other; the bits sent must match timing derived from a pre-agreed baud rate.
  - Full-duplex: lines are TX and RX which are separate (transmit - receive) (allows for separate communication)
  - Frame-based (data is sent in frames wrapped in start/stop bits allowing receiver to detect boundaries w/o clock)
- Saying UART typically refers to the hardware peripheral (chip / IP) converting parallel data from CPU into a serial bitstream, or the other way around
- Some UART implementations add hardware flow control lines:
  - RTS and CTS which are used to prevent buffer overruns
  - Buffer overruns happen when one device cannot keep up with the other

- UART Frame:
  - Since there is no clock line, UART is entirely frame-based. Start/stops are transmitted as voltage levels on the wire using a fixed frame format
  - A single frame:
    - Idle (line sits HIGH)
    - Start bit (line is pulled low for a single bit period)
    - Data bits (5-9, most commonly 8, w/ the LSB sent first)
    - Parity bit (error-check bit that can be even, odd, or none)
    - Stop bit (line returns HIGH for 1 or 2 bit periods, marking end of the frame)
  - Frames are often written in shorthand: 8N1 - 8 data, no parity, 1 stop

- Baud rate is the bits per second